

Comprehensive scrutinization of ternary hybrid Casson nanofluid flow in a conducting porous rotating disk with internal heating

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The utilization of a ternary hybrid nanofluid, a recent development in the realm of nanofluids, can result in improved heat transfer. In the ongoing study, a ternary hybrid nanofluid flow is utilized, and it is carried out atop a porous spinning disc, which is exposed to a magnetic field, heat generation, thermal radiation, and Casson fluid. In this study, Blood/Water are taken as base fluids and (Ag–Au–Al₂O₃) are considered as a ternary hybrid nanoparticle. Nanoparticles made of gold and silver are put to use in a vast number of industries and fields, including nanotechnology and medicine. As a result of surface effects and quantum effects, these precious metals exhibit unique features in nanoform that play a vital role in with optical, magnetic, chemical, and mechanical behavior. Al₂O₃ unique optical, physical and biochemical qualities make it worthwhile for numerous uses, including nanophotonic, catalysis and the fabrication of high-energy composites. A set of relevant similarity transformations is used to generate non-dimensional forms of controlling paired nonlinear Partial Differential Equations (PDEs). MATLAB is used to perform a numerical solution with ODE45. In addition, the velocity outline decreases, and the temperature profile increases slightly before decreasing over a revolving disk when the values of magnetic parameters are increased. The distribution and radiant heat components heat up as the level gets higher.

Aim and objectives of the study:

The aim of this study is to analyze the Comprehensive Scrutinization of Ternary hybrid and Casson flow in a conducting porous rotating disk with internal heating. The primary objective of this analysis is to increase awareness of the impending energy crisis among those working in the industrial and technological sectors. Ternary nanoparticles (NPs) have a wide range of applications, which lends credence to the developed model. For example, Al₂O₃ can be used in a variety of ways that benefit society, it is used in water purification to remove water from the gas streams and extend people's lives.

Keywords: Ternary hybrid nanofluid, radiation; MHD; porous medium; heat generation; Casson model.

Nomenclature

T_{∞}	Ambient fluid temperature (K)
f	Base fluid
$THnf$	Ternary hybrid nanofluid
β	Casson parameter
(u, v, w)	Velocity elements
(r, ϕ, z)	Directions, m/s
μ	Dynamic viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)
ρ	Density (kg m^{-3})
σ	Electric conductivity (Sm^{-1})
$\text{Ec} = \frac{r^2 \Omega^2}{(C_p)_f (T_f - T_{\infty})}$	Eckert number
c_p	Heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)
h_f	Heat transfer coefficient, $\text{W/m}^2/\text{k}$
$Q = \frac{Q_0}{\Omega(\rho C_p)_f}$	Heat absorption
ν	Kinematic viscosity ($\text{m}^2 \text{s}^{-1}$)
P	Pressure, (pa)
$K = \frac{\nu_f}{\Omega K^*}$	Porosity parameter
ϕ_1, ϕ_2, ϕ_3	Nanoparticles volume fractions

$Rd = \frac{4\sigma^* T_w^3}{k^* k_f}$	Radiation parameter
$Re = r \left(\frac{r\Omega}{v_f} \right)$	Reynolds number
T_w	Surface temperature, K
$\gamma = \frac{S}{\Omega}$	Stretching ratio parameter
E	Strength of electric field (N/C)
σ^*	Stefan Boltzmann constant
T_f	Temperature of heated fluid, K
k	Thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
β	Thermal expansion K^{-1}
α_f	Thermal diffusivity, m^2/s
k^*	Mean absorption coefficient
$M = \frac{\sigma B_0^2}{\Omega \rho_f}$	Magnetic interaction parameter

1. Introduction

Thermal engineering, energy processes and industrial processes rely on effective energy transmission in liquid mechanics with a rotatory disc shape. These include aircraft engines, radiators, electric-producing power plants and manufacturing. Many areas of aeronautical science and engineering make use of the fluid flow created by a revolving disc, including but not limited to the following: Rotational machinery, heating and cooling, gadgets for computers, spinning fans employed in thermal power, healthcare equipment, and numerous others are examples of some of the applications of thermal energy. Karman¹ studied the fluid flow brought on by a rotating disc. Scientists worldwide are looking into disc rotation as a geometry using different models of fluids because of its potentially life-changing applications. Several recent theoretical and experimental studies by Ramzan *et al.*,² Farooq *et al.*,³ Miklavčič and Wang⁴ have explored various flow elements in a rotating disk. Recently, quantitative research on hybrid nanofluids has been performed to learn more about their heat-transfer properties and the amount to which this new classification of nanofluids achieves radiative impact. Colloids, which are made up of two or more solid particles of micron size or less, are prime examples of mixed nanofluids that mix with particles. Because of the importance of these applications, a growing number of scientists are exploring using mixed nanofluids to address practical heat transfer challenges. Optical, thermos-physical, morphological and rheological qualities can be improved by merging two or more nano-particle types in a single base liquid to create a hybrid nanofluid. Hybrid nanofluids are expected to replace basic nanofluids for a variety of applications, with the exception of those requiring a wider range of absorption, a lower rate of elimination, excellent thermal conductivity, a little pressure drop, a modest amount of frictional loss and pumping power will be low. Bringing together the thermo-physical characteristics of highly effective nanoparticles can

improve the heat transmission qualities of a base liquid. Recent developments in hybrid nanomaterials are done by the researchers Alkuhayli,⁵ Gul *et al.*,⁶ Basit *et al.*,⁷ Farooq.⁸ Ternary hybrid nanofluids will enhance heat transfer rate and individual nanoparticles' distinct characteristics. Currently, tri-hybrid facilities may be found in use in the following areas: solar procedures, heat transfer, automobile production, photovoltaic thermal collector, electrical chillers, nuclear networks, broadcasting, cruise and bioengineering. Ternary Hybrid nanofluids have been the subject of some researches, and this area has been illuminated by the works cited in Shamshuddin,⁹ Sarangi *et al.*¹⁰ and Alqawasmi *et al.*¹¹

Magnetohydrodynamics (MHD) is an outstanding topic of this study, and its concepts are well known to scientists who study solar physics, high-energy physics or stars. The fluid's electromagnetic and hydrodynamic behaviors interact to produce this effect. Magnetic field dynamics in electrically conducting fluids like liquids and metal plasmas are the focus of this physical framework. Magnet is a word that connotes magnetism, while hydro refers to water. Examples of how MHD can be put to use include magnetic flux pipes, arcs, filaments and loops; hydromagnetic dynamo movement; turbulence; and magnetic-convection. Electromagnetic impact, liquid-metal nuclear reactors cooling, and the plasmas behavior in fusion reactors are all examples of how MHD is applied in the engineering area. Magnetic fields are used in aircraft, chemistry and astrophysics, among other professions. The strength and distribution of the provided metamaterials affect the flow properties. The literature on MHD flow of various fluid models is extensive. Many researchers in the field of fluid mechanics have conducted experiments to better understand the influence of a magnetic field on the movement of fluids.^{12–16} Thermal radiation (TR) is a kind of electromagnetic radiation that is produced when nanoparticles in materials move as a consequence of their temperature changing. The process of TR is the basis for the transformation of thermal energy into electromagnetic energy. Surface conditions are directly related to the TR characteristics. The Rosseland flow which is linear, including heat radiation and energy transport, is studied by a number of researchers.^{17–20} When it comes to regulating heat flow, TR delivers notable results.

Motivated by the literature, the purpose of this research is to apply the Casson model to the motion of a radiatively spinning disc through which a tri-hybrid nanofluid flows. MHD, heat source/sink and porosity are considered. (Ag–Au–Al₂O₃) are the three hybrid nanoparticles and Blood/Water are taken as base fluids. Silver (Ag) nanoparticles are significant in food because they are antibiotic and anti-fungicidal. Manufacturing, pastes, polymer compounds, nourishment and detergents use it. Connectors for electrical equipment and electronic components can be fabricated from gold (Au) since it is an excellent conductor of both electricity and heat and remains shiny when exposed to the air. Spacecraft and tall buildings can benefit from its ability to reflect infrared radiation because of its application in solar heat protection. It is possible to treat cancer with a radioactive gold isotope. Aluminium oxide is an excellent ceramic oxide that is widely used in the synthesis of adsorbents and catalysts. It is also utilized to make many of chemicals with significant economic value and in the

aerospace sector. With the help of the MATLAB software, the Ordinary Differential Equations (ODEs) are solved using the ODE45 method. The effects of each variable on the displayed profiles are shown. The results can be used in catalysis, which can be used in autos as well as other types of electrical equipment. By taking nanoparticles into account which conduct more heat, the heat transfer is substantially higher in comparison to other systems, making evaluation of this model vital for the movement of heat.

2. Mathematical Modeling

In this study, we investigate a steady incompressible Thnf (Ternary hybrid nano-fluid) flow across a porous revolving disk, when exposed to magnetic field, heat generation, TR and Casson fluid. The disc maintains a steady angular velocity at $z = 0$ throughout its rotation. Figure 1 clearly shows that the velocities (u, v, w) are increasing in (r, ϕ, z) directions. In this research, we focus on a ternary hybrid nanoparticle (Ag–Au–Al₂O₃), where blood and water serve as the basis fluids.

The governing equations are developed by using the studies of Nayak *et al.*,²¹ Shaw *et al.*²² and Arafa *et al.*²³

$$\frac{\partial u}{\partial r} + \frac{u}{r} + \frac{\partial w}{\partial z} = 0, \quad (1)$$

$$u \frac{\partial u}{\partial r} - \frac{v^2}{r} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho_{Thnf}} \frac{\partial p}{\partial r} + \left(1 + \frac{1}{\beta}\right) \frac{\mu_{Thnf}}{\rho_{Thnf}} \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} - \frac{u}{r^2} + \frac{\partial^2 u}{\partial z^2}\right) - \frac{\mu_{Thnf}}{\rho_{Thnf}} \frac{u}{K^*} - \frac{\sigma_{Thnf}}{\rho_{Thnf}} (B^2 u), \quad (2)$$

$$u \frac{\partial v}{\partial r} + \frac{uv}{r} + w \frac{\partial v}{\partial z} = -\frac{1}{\rho_{Thnf}} \frac{\partial p}{\partial \phi} + \left(1 + \frac{1}{\beta}\right) \frac{\mu_{Thnf}}{\rho_{Thnf}} \left(\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} - \frac{v}{r^2} + \frac{\partial^2 v}{\partial z^2}\right) - \frac{\mu_{Thnf}}{\rho_{Thnf}} \frac{v}{K^*} - \frac{\sigma_{Thnf}}{\rho_{Thnf}} (B^2 v), \quad (3)$$

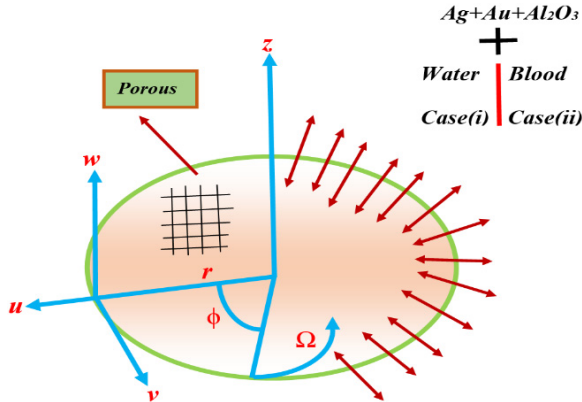


Fig. 1. (Color online) Flow justification of the ternary hybrid nanofluid.

$$u \frac{\partial w}{\partial r} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho_{Thnf}} \frac{\partial p}{\partial z} + \left(1 + \frac{1}{\beta}\right) \frac{\mu_{Thnf}}{\rho_{Thnf}} \left(\frac{\partial^2 w}{\partial r^2} + \frac{1}{r} \frac{\partial w}{\partial r} + \frac{\partial^2 w}{\partial z^2} \right), \quad (4)$$

$$u \frac{\partial T}{\partial r} + w \frac{\partial T}{\partial z} = \frac{k_{Thnf}}{(\rho c_p)_{Thnf}} \left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial z^2} \right) + \frac{16\sigma^* T_1^3}{3k^*(\rho c_p)_{Thnf}} \\ \times \left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial z^2} \right) + \frac{Q_0}{(\rho c_p)_{Thnf}} (T - T_\infty) + \frac{\sigma_{Thnf}}{(\rho c_p)_{Thnf}} (B^2 u^2) \quad (5)$$

with

$$u = rS, \quad v = r\Omega, \quad w = 0, \quad T = T_w \quad \text{at } z = 0, \\ u \rightarrow 0, \quad v \rightarrow 0, \quad T \rightarrow T_\infty, \quad \text{at } z \rightarrow \infty. \quad (6)$$

The following are some mathematical models of the thermophysical features of Ternary hybrid nanofluid:

$$A_1 = \frac{\mu_{Thnf}}{\mu_f}, \quad A_2 = \frac{\rho_{Thnf}}{\rho_f}, \quad A_3 = \frac{\sigma_{Thnf}}{\sigma_f}, \quad A_4 = \frac{(\rho c_p)_{Thnf}}{(\rho c_p)_f}, \quad A_5 = \frac{k_{Thnf}}{k_f}. \quad (7)$$

The suitable self-similarity variables are

$$(u, v, w) = (r\Omega f'(\zeta), r\Omega g(\zeta), -\sqrt{2\Omega} v_f f(\zeta)), \\ (p, T) = (p_\infty - \Omega \mu_f p(\zeta), T_\infty + (T_f - T_\infty)\theta), \zeta = \sqrt{\frac{2\Omega}{v_f}} z. \quad (8)$$

The thermophysical characteristics of ternary hybrid nanofluids have been studied by Kumar *et al.*^{24,25}

The ternary hybrid nanofluid viscosity model is

$$\mu_{Thnf} = \frac{\mu_f}{(1 - \varphi_1)^{2.5} (1 - \varphi_2)^{2.5} (1 - \varphi_3)^{2.5}}.$$

The ternary hybrid nanofluid density model is

$$\frac{\rho_{Thnf}}{\rho_f} = (1 - \varphi_1) \left\{ (1 - \varphi_2) \left[(1 - \varphi_3) + \frac{\varphi_3 \rho_{sp3}}{\rho_f} \right] + \frac{\varphi_2 \rho_{sp2}}{\rho_f} \right\} + \frac{\varphi_1 \rho_{sp1}}{\rho_f}.$$

Ternary hybrid nanofluid Electric conductivity is given by

$$\frac{(\sigma)_{Thnf}}{(\sigma)_f} = \varphi_1 \frac{(\sigma)_{sp1}}{(\sigma)_f} + (1 - \varphi_1) \left[(1 - \varphi_1) \left\{ (1 - \varphi_3) + \varphi_3 \frac{(\sigma)_{sp3}}{(\sigma)_f} \right\} + \varphi_2 \frac{(\sigma)_{sp2}}{(\sigma)_f} \right].$$

Ternary hybrid nanofluid specific heat structure is given by

$$\frac{(\rho c_p)_{Thnf}}{(\rho c_p)_f} = \varphi_1 \frac{(\rho c_p)_{sp1}}{(\rho c_p)_f} + (1 - \varphi_1) \left[(1 - \varphi_1) \left\{ (1 - \varphi_3) + \varphi_3 \frac{(\rho c_p)_{sp3}}{(\rho c_p)_f} \right\} + \varphi_2 \frac{(\rho c_p)_{sp2}}{(\rho c_p)_f} \right].$$

In his model, heat conduction in ternary hybrid nanofluids is given by

$$\left. \begin{aligned} \frac{k_{Thnf}}{k_{hnf}} &= \frac{(k_3 + 2k_{hnf} - 2\varphi_3(k_{hnf} - k_3))}{(k_3 + 2k_{hnf} + \varphi_3(k_{hnf} - k_3))} \\ \frac{k_{hnf}}{k_{nf}} &= \frac{(k_2 + 2k_{nf} - 2\varphi_2(k_{nf} - k_2))}{(k_2 + 2k_{nf} + \varphi_2(k_{nf} - k_2))} \\ \frac{k_{nf}}{k_f} &= \frac{(k_1 + 2k_f - 2\varphi_1(k_f - k_1))}{(k_1 + 2k_f + \varphi_1(k_f - k_1))} \end{aligned} \right\}.$$

Now, by plugging in (7) and (8) into Eqs. (2)–(6), we get the following form:

$$\frac{A_1}{A_2} 2 \left(1 + \frac{1}{\beta} \right) f''' - \frac{A_1}{A_2} K f' + 2f f'' - f'^2 + g^2 - \frac{A_3}{A_2} M f' = 0, \quad (9)$$

$$\frac{A_1}{A_2} 2 \left(1 + \frac{1}{\beta} \right) g'' - \frac{A_1}{A_2} K g + 2f g' - 2f' g - \frac{A_3}{A_2} M g = 0, \quad (10)$$

$$\frac{1}{A_4} \left(A_5 + \frac{4}{3} R d \right) \theta'' + \text{Pr} \left[f \theta' + \frac{Q \theta}{2A_4} + \frac{A_3}{A_4} M E c (f')^2 \right] = 0. \quad (11)$$

Non-dimensional boundary conditions are as follows:

$$\begin{aligned} f' = \gamma, \quad g = 1, \quad f = 0, \quad \theta' = 1 \quad \text{at } \zeta = 0, \\ f' \rightarrow 0, \quad g \rightarrow 0, \quad \theta \rightarrow 0, \quad \text{as } \zeta \rightarrow \infty. \end{aligned} \quad (12)$$

The dimensional forms of C_f and Nu_r are given by

$$C_f = \frac{\sqrt{\tau_r^2 + \tau_\varphi^2}}{\rho_f (r\Omega)^2}, \quad (13)$$

$$\text{Nu}_r = - \left(k_{Thnf} + \frac{16\sigma^* T_1^3}{3k_f k^*} \right) \frac{r \left(\frac{\partial T}{\partial z} \right)_{z=0}}{(T_f - T_\infty)}, \quad (14)$$

where

$$\tau_r = \mu_{Thnf} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} \right)_{z=0} = \mu_{Thnf} (r\Omega) \sqrt{\frac{2\Omega}{v_f}} F''(0), \quad (15)$$

$$\tau_\varphi = \mu_{Thnf} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial r} \right)_{z=0} = \mu_{Thnf} (r\Omega) \sqrt{\frac{2\Omega}{v_f}} G'(0). \quad (16)$$

The non-dimensional forms of Eqs. (13) and (14) are converted as

$$(\text{Re}_r)^{1/2} C_f = \sqrt{2} A_1 \sqrt{[F''(0)]^2 + [G'(0)]^2}, \quad (17)$$

$$(\text{Re}_r)^{-1/2} \text{Nu}_r = -\sqrt{2} \left(\frac{k_{\text{Thnf}}}{k_f} + \frac{4}{3} \text{Rd} \right) \theta'(0). \quad (18)$$

3. Numerical Procedure

Considering the circumstances (12), the modified equations (9)–(11) are solved using the built-in MATLAB function ODE45. Since it is a built-in function, we use the ODE45 solver to resolve this issue.³⁰

We combined the ODE45 solver to adjust for this.

$$\text{Sol} = \text{solver}(\text{odefun}, [t_0, t_f], y_0 \dots).$$

We make the following assumptions before we start the coding process.

$$f = h_1, f' = h_2, f'' = h_3, g = h_4, g' = h_5, \theta = h_6, \theta' = h_7.$$

4. Results and Discussion

The collection of Partial Differential Equations (PDEs) are represented by Eqs. (1)–(3) and the corresponding boundary conditions (4) are converted into a set of ODEs (10)–(11) and Eq. (12) and addressed by using the MATLAB. The physical nature of the problem is made clearer by discussing the outcomes of various parameters and illustrating the results in this section. Magnetic parameter M , porosity K , non-Newtonian fluid parameter, i.e. Casson parameter β , Q and Rd were realized. The following analysis considers how these factors influence the temperatures, skin friction and Nusselt numbers found throughout the various velocity, temperature and Nusselt number distributions. Table 1 shows the thermophysical properties of the base fluids and tri-hybrid nanoparticles.

4.1. Illustration of velocity profiles

The behavior of the magnetic parameter M in the radial and azimuthal velocity sketch is shown in Figs. 2 and 3, respectively. The radial and azimuthal velocity profiles for (Ag–Au–Al₂O₃) Nanofluid with Blood/Water as base fluid $f'(\zeta)$ declined as the M increased. Both velocity profiles slant because of the external resistance

Table 1. Thermophysical characteristic of Ternary hybrid nanoparticles using the study of Ullah *et al.*²⁶ and Basit *et al.*²⁷

Property	Blood	Water	Au	Ag	Al ₂ O ₃
Density ρ (kgm ⁻³)	1063	997	19300	10500	8933
Specific heat C_p (Jkg ⁻¹ K ⁻¹)	3594	4180	129.1	235	385
Heat conductivity k_f (Wm ⁻¹ K ⁻¹)	0.492	0.6071	318	428	401
Electrical conductivity σ (Ωm) ⁻¹	6.67×10^{-1}	0.005	4.52×10^7	25000	5.96×10^7
Pr	21	6.2			

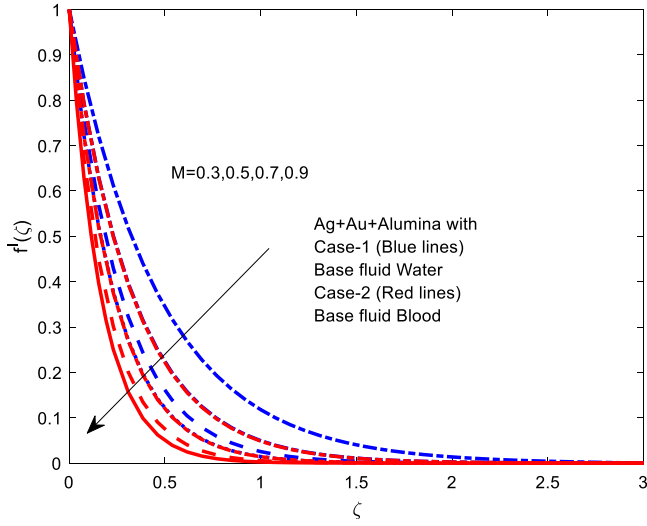


Fig. 2. (Color online) Influence of M on radial velocity.

introduced by an enlarging M . Physically, the velocity gradients are slowed down, which makes the momentum barrier layer thinner and slows the flow. Lorentzian magnetohydrodynamic drag, created by the interaction of the magnetic field, opposes the flow of the fluid throughout the entire domain. When an oblique magnetic field is applied to semiconducting nanoparticles, a Lorentz force is generated. Additionally, when the intensity of the magnetic field is raised, the Lorentz force also rises. This reduces boundary layer viscosity at high speeds by opposing fluid flow.

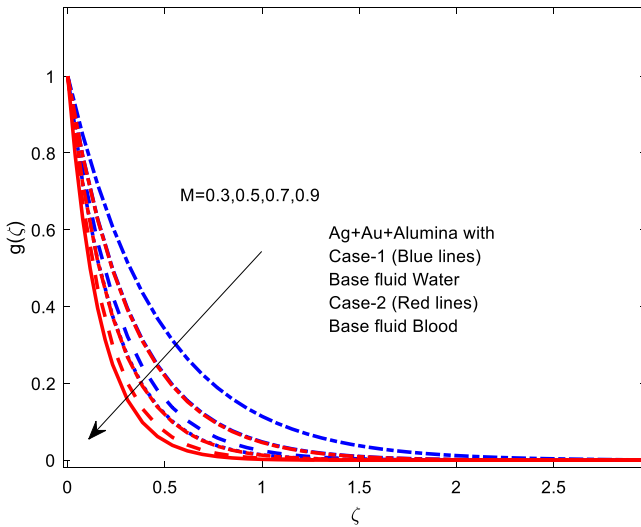


Fig. 3. (Color online) Influence of M on azimuthal velocity.

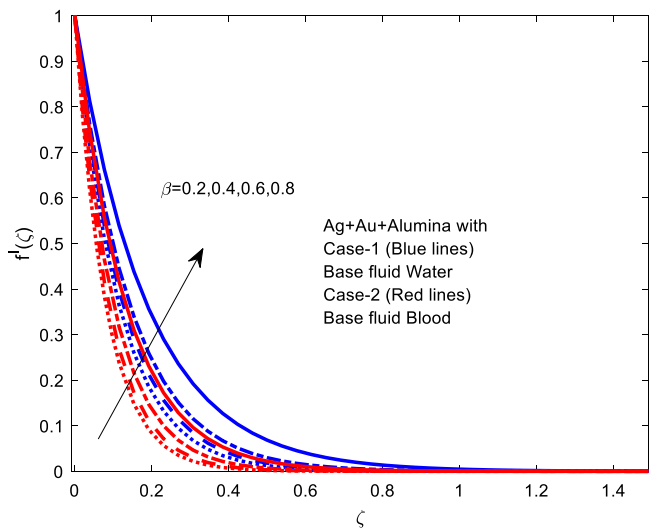


Fig. 4. (Color online) Influence of β on radial velocity.

The Casson fluid factor β has an effect on the radial and azimuthal velocity curves are seen in Figs. 4 and 5. A rise in the Casson fluid parameter is responsible for the expanding velocity, which is a consequence of the increase, and as a straight consequence, the viscosity of the boundary layer is growing as well. Physically, Casson fluid is having shear thinning characteristic implicating lowering of viscosity thereby enhancing the fluid velocity. On comparison, $g(\zeta)$ magnitude is maximum for ternary hybrid nanofluid with water as the base fluid than blood.

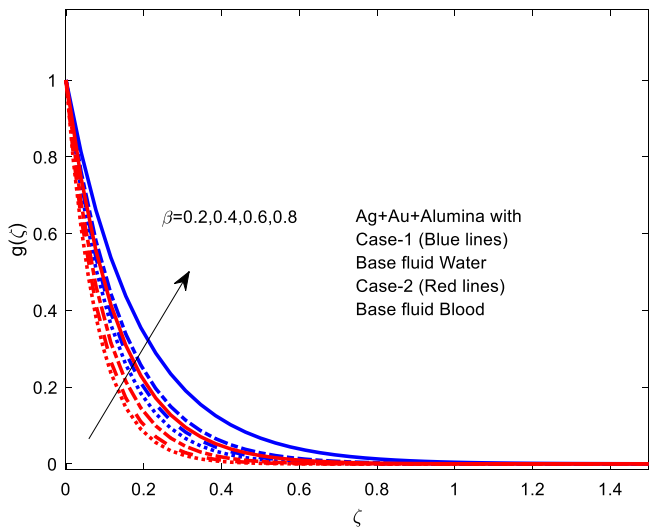


Fig. 5. (Color online) Influence of β on azimuthal velocity.

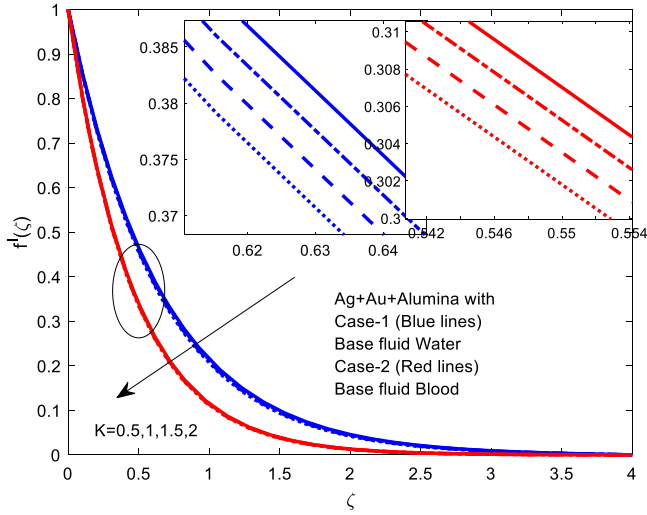


Fig. 6. (Color online) Influence of K on radial velocity.

Figures 6 and 7 illustrate in greater detail the effect that the $f'(\zeta)$ and $g(\zeta)$ velocity outlines have on the fluctuation of the porosity parameter K . It would appear that the porosity parameter lowers itself when the velocity components are considered. Because increasing the quantity of the K increases the thickness of the boundary layer to a greater extent, which in turn causes the velocity to decrease. Physically, we know that the porosity parameter acts to restrict the motion of the fluid on the outer layer of the disk. This is due to the fact that the development in the

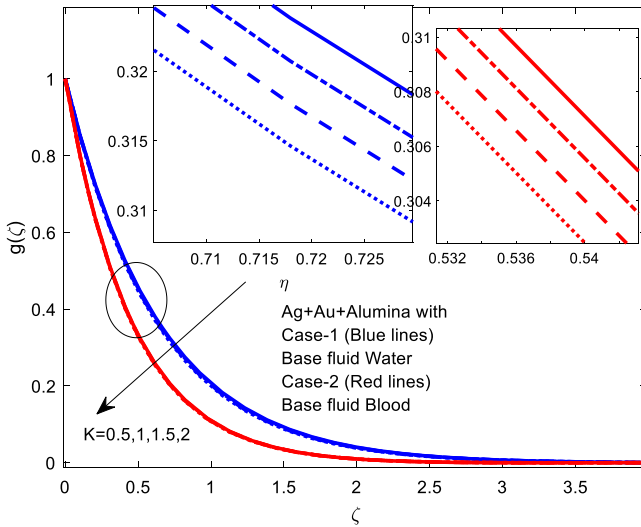


Fig. 7. (Color online) Influence of K on azimuthal velocity.

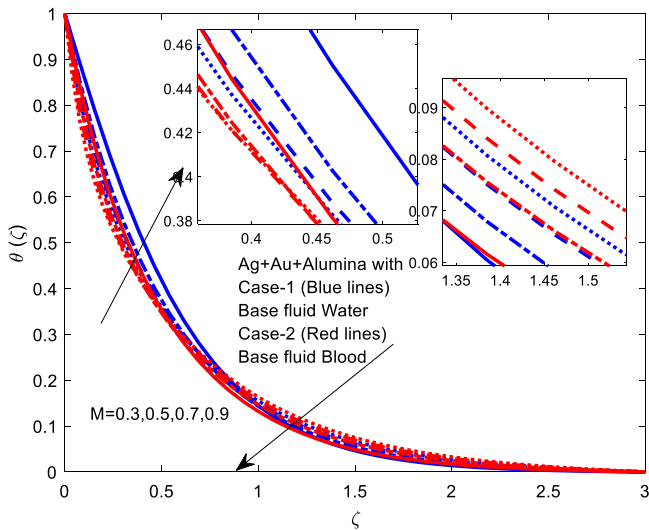


Fig. 8. (Color online) Influence of M on temperature.

porous material acts as a barrier to the passage of fluid, which in turn causes flow retardation.

4.2. Illustration of temperature profiles

The influence of (M) on the temperature profile is depicted in Fig. 8. Physically, thermal energy strengthens the Lorentz force by motivating hybrid nanofluids to dissipate sub kinetic energy. As a result of the fluid’s relatively slow motion, the kinetic energy is converted into heat energy, it leads to a rise in temperature for the area as a

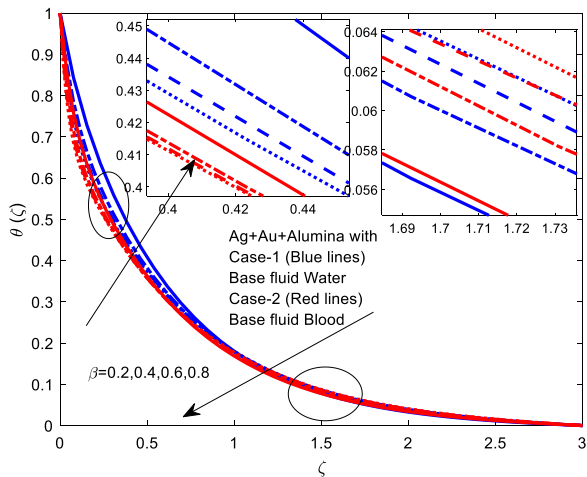


Fig. 9. (Color online) Influence of β on temperature.

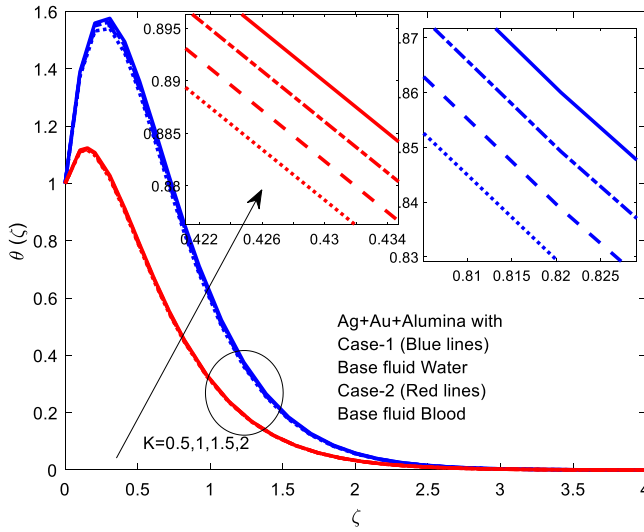


Fig. 10. (Color online) Influence of K on temperature.

whole close to the boundary layer. In addition to this, it has been shown that an enhancement in the strength of the magnetic field causes both the thermal and momentum boundary layers of the hybrid nanofluid to grow smaller. The boundary layer temperature rises because the size of the velocity profiles decreases with increasing magnetic factor. The density of the hybrid nanofluid is shown to be inversely proportional to the magnetic parameter. Increases in the magnetic field component were controlled to higher temperatures because they reduce the fluid density.

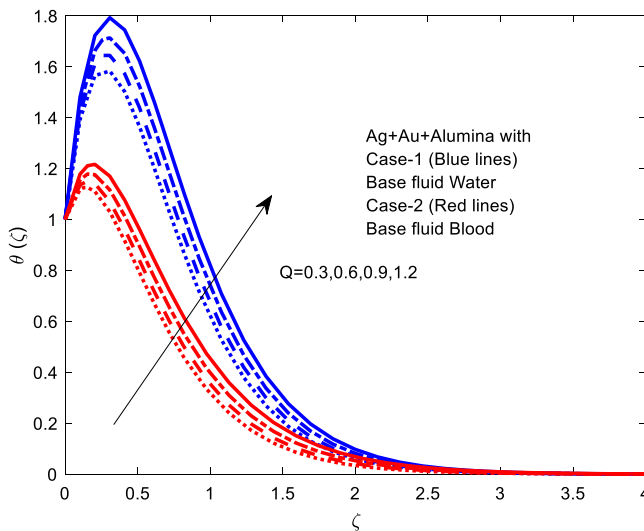


Fig. 11. (Color online) Influence of Q on temperature.

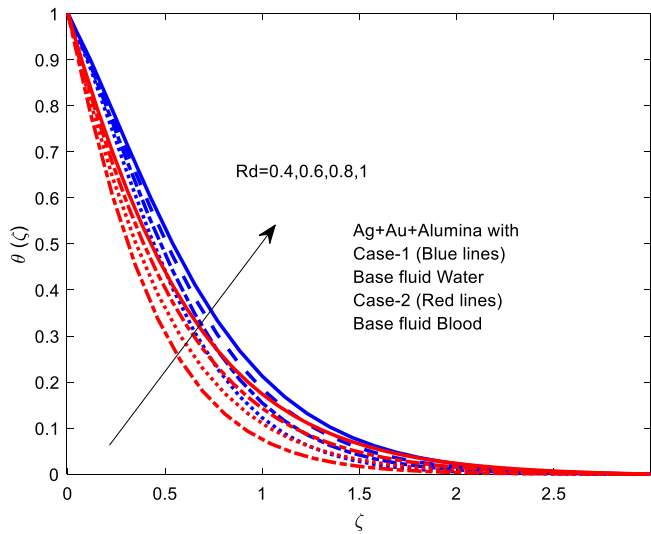


Fig. 12. (Color online) Influences of Rd on temperature.

As far as temperature distribution is concerned, rise in β gives rise to the amplification of fluid temperature of ternary hybrid nanofluid in the entire boundary layer region (Fig. 9). Indeed, rise in Casson parameter implicates the more thinning nature of the fluid. Therefore, viscosity of the fluid reduces thereby velocity of the fluid enhances. Consequently, the kinetic energy of the molecules taking part in the collision process increases. As a result, the temperature of the fluid ameliorates. It is visualized that $\theta(\zeta)$ is greater near the solid boundary and it gradually peters out as we move away from the boundary. Figure 10 shows how changing the porosity factor

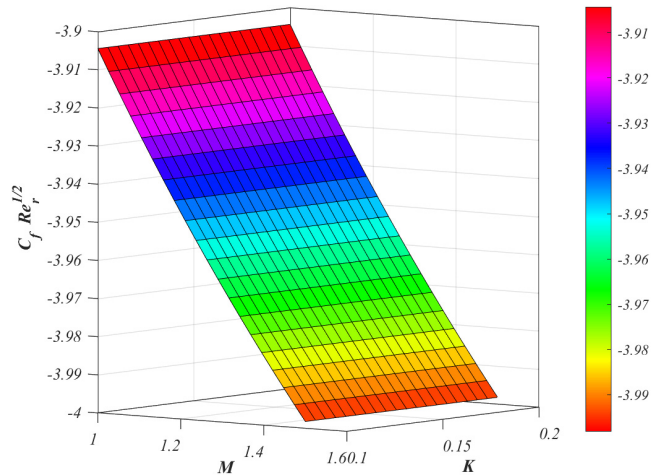


Fig. 13. (Color online) Influence of M and K on $C_f Re_r^{1/2}$.

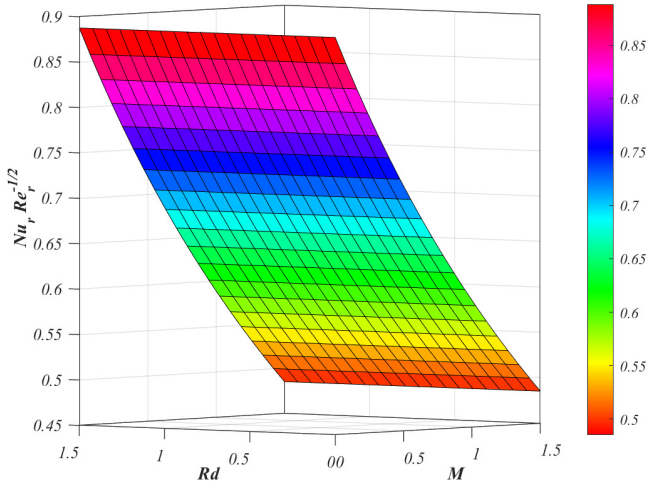


Fig. 14. (Color online) Influence of Rd and M on $Nu_r Re_r^{-1/2}$.

K affects the temperature pattern. Increasing the porosity parameter's permeability causes the fluid temperature to rise. The raise in temperature is more for water than the blood. In Fig. 11, we have a look at how the influence of the fluid temperature profile changes depending on the value of the heat generation parameter Q . It demonstrates that there is a rise in the temperature profile as the value of the heat generation Q grows. Physically, larger levels of Q increase the inherent energy of liquid particles, causing temperature to rise. The temperature increase is more for water than the blood due to the viscosity. Figure 12 depicts the effect of various

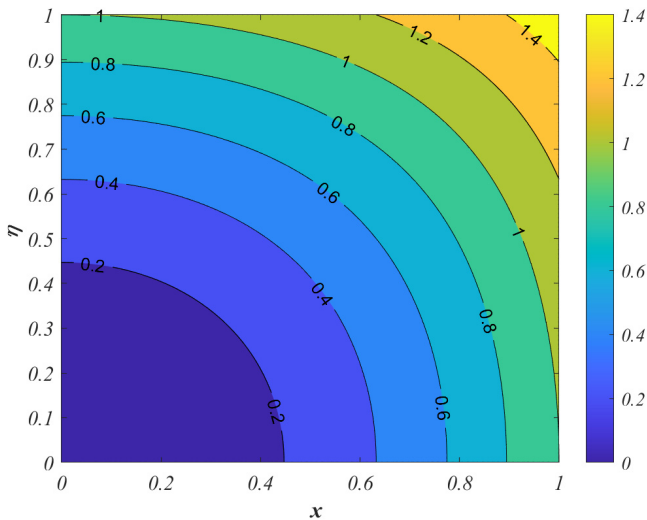


Fig. 15. (Color online) Stream lines for magnetic field $M = 0$.

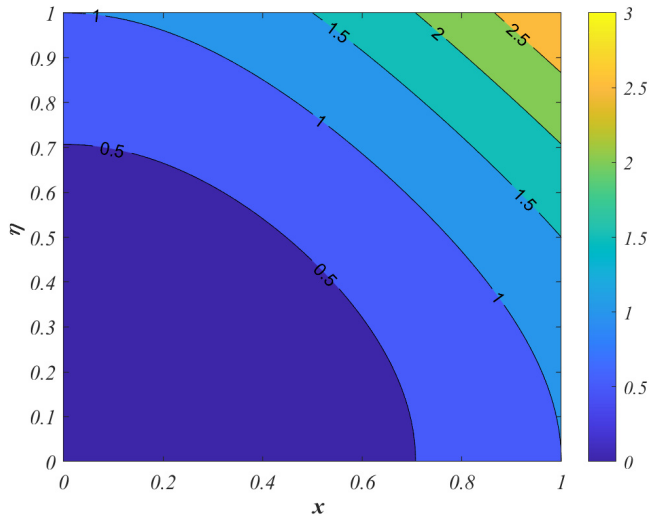


Fig. 16. (Color online) Stream lines for magnetic field $M = 0.2$.

radiation parameter (Rd) values on the temperature of hybrid nanofluid. TR has a positive physical effect on the medium’s thermal diffusibility, which increases the temperature profile. Physically, higher temperature and a thicker thermal boundary layer are correlated with increased TR parameters. Figures 13 and 14 demonstrate the (M & K), and (Rd & M) influences on $C_f Re_r^{1/2}$ and $Nu_r Re_r^{-1/2}$. For the higher values of both cases, both profiles are enhanced due to the high range of Lorentz force

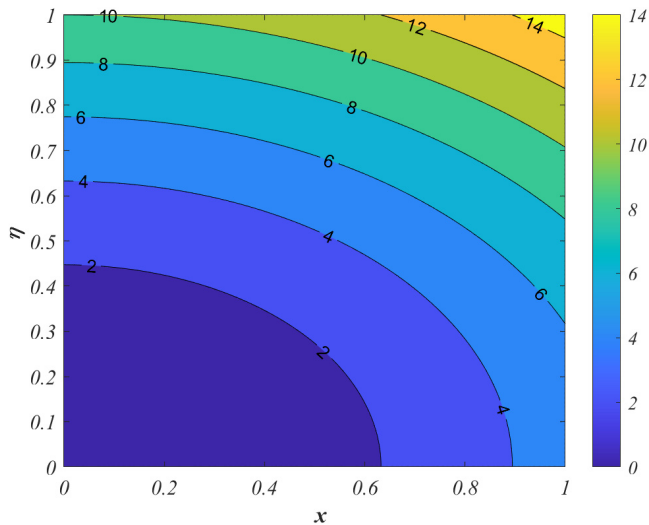


Fig. 17. (Color online) Stream lines for magnetic field $M = 0.4$.

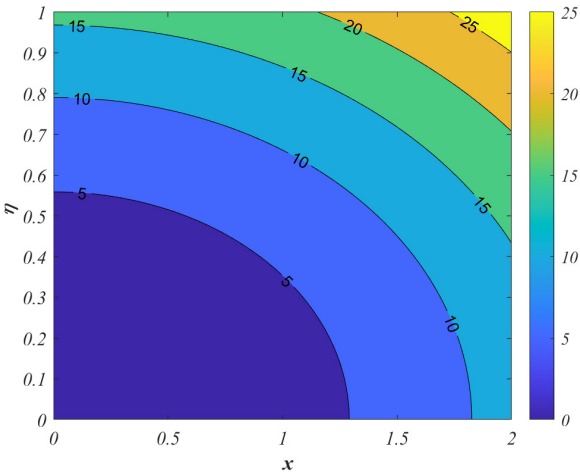


Fig. 18. (Color online) Stream lines for magnetic field $M = 0.6$.

which goes to the main stream to the fluid flow. Figures 15–18 demonstrate the $M = 0, 0.2, 0.4, 0.6$ influences on streamlines. The nanoparticle magnetic field strength attractions to the magnetic nature molecules more towards the top.

Variations in Skin friction and Nusselt number for various values of M, β, Rd , and K are given in Table 2. It is clear from Table 2 that the Skin friction and Nusselt number increases with growing values of Magnetic parameter, Casson parameter,

Table 2. The physical parameter values of the skin friction and Nusselt number.

M	β	Rd	K	Skin friction		Nusselt number	
				Case –I	Case –II	Case –I	Case –II
0.3				2.409287	2.334201	2.861743	2.774573
0.5				3.104293	3.007226	3.674706	3.561852
0.7				3.669968	3.555049	4.337788	4.204089
0.9				4.159413	4.029065	4.912124	4.760414
	0.2			1.901927	1.843634	2.295360	2.228096
	0.4			2.490216	2.413891	3.006067	2.918028
	0.6			2.852904	2.765464	3.444180	3.343331
	0.8			3.105850	3.010658	3.749712	3.639927
		0.4		1.404675	1.361628	1.694536	1.644835
		0.6		1.411609	1.368780	1.702243	1.652778
		0.8		1.418510	1.375895	1.709930	1.660700
		1.0		1.425378	1.382974	1.717597	1.668599
			0.5	1.404675	1.361628	1.694536	1.644835
			1.0	1.411609	1.361628	1.702243	1.644835
			1.5	1.418510	1.361628	1.709930	1.644835
			2.0	1.425378	1.361628	1.717597	1.644835

Table 3. Comparative values of the skin friction coefficient.

K	Skin friction		Skin friction
	Sekhar <i>et al.</i> ²⁸	Sekhar and Reddy ²⁹	Present study
0	0.30051	0.300512	0.300510
0.5	0.26436	0.264361	0.264363
1	0.23697	0.236969	0.236968

radiation parameter and porosity parameters for both the cases (Ag–Au–Al₂O₃) with water and (Ag–Au–Al₂O₃) with blood. Table 3 shows the validation of the present solutions with already available solution under limited case. It is found that the skin friction values are exactly matched with available solutions Sekhar *et al.*²⁸ and Sekhar and Reddy.²⁹

5. Conclusions

Some specific applications of gold and silver nanoparticles include silver and gold nanoparticles can be synthesized at the nanoscale and have unique biological properties like antibacterial, antifungal, antiviral, antiparasitic, antiplatelet anti-inflammatory, Drug delivery and anti-tumor activity. The mathematical investigation of tri-hybrid nanofluid movement moving along a radiatively rotating disk by using the Casson model was discussed. It is immersed in a porous material and is affected by heat sink/source and magnetic effects. Silver, gold and Al₂O₃ nanoparticles are found in base fluids such as water and blood. The main conclusions of the study can be summarized as follows:

- The radial and azimuthal velocity profiles for tri-hybrid nanofluid declined as the magnetic parameter *M* and porosity parameter *K* increased.
- Increasing the magnetic field parameter, Casson parameter, heat generation and porosity parameter causes the fluid temperature increases.
- TR has a positive physical effect, which increases the temperature profile.
- The Skin friction and Nusselt number increase with growing values of Magnetic parameter, Casson parameter, radiation parameter and porosity parameters for both the cases (Ag–Au–Al₂O₃) with water and (Ag–Au–Al₂O₃) with blood.

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